

Sangola College, Sangola
Subject: Object Oriented Programming by using C++
Class: B.Sc (ECS)-I

ASSIGNMENT -1

- 1) Write a program which find sum of digits (digit sum) of entered number.
- 2) Write a program which check entered number is Perfect or not.
- 3) Write a program which finds face value of entered number.
- 4) Write a program which reverses the entered number.
- 5) Write a program which check entered number is Palindrome or not.
- 6) Write a program which check entered number is Armstrong or not.
- 7) Write a program which check entered number is Prime or not.
- 8) Write a program which finds factorial of an entered number.
- 9) Write a program which prints Fibonacci series up to 'n' numbers.
- 10) Write a program which prints first 20 even numbers.
- 11) Write a program which prints first 30 palindrome numbers.
- 12) Write a program which finds Armstrong numbers between 1 to 1000.
- 13) Write a program which finds Prime numbers between 1 to 1000.
- 14) Write a menu driven program which sort array elements either in ascending Order or in descending order.
- 15) Write a program which check entered string is Palindrome or Not.
- 16) Write a program which print diagonal elements of matrix.
- 17) Write a program which find sum of elements of matrix.

- 18) Write a program which finds Multiplication of Two matrices.
- 19) Write a program which find maximum and minimum number of the matrix.
- 20) Write a program which Check entered number is Magic or not

ASSINMENT-2

- 1) Write the programs which prints given output:

I) <pre> 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5 </pre>	II) <pre> 5 4 3 2 1 5 4 3 2 5 4 3 5 4 5 </pre>
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III)

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          A
        A  B
      A  B  C
    A  B  C  D
  A  B  C  D  E

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IV)

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A  B  C  D  C  B  A
A  B  C      C  B  A
A  B          B  A
A                      A

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V)	1	2	3	4	5
	1	2	3	4	
	1	2	3		
	1	2			
	1				
	1	2			
	1	2	3		
	1	2	3	4	
	1	2	3	4	5

VI)	1			
	2	3		
	4	5	6	
	7	8	9	10

VII)	1			
	0	1		
	0	1	0	
	1	0	1	0

ASSIGNMENT-3

- 1) Write a program to implement MSEB class with appropriate data members and member functions that calculates light bill.
- 2) Write a program to generate pay slip of an employee. Take appropriate data members and member functions.
- 3) Write a program which prints student mark sheet by taking name, address, class, division and marks of six subjects as input.
- 4) Write a program to calculate overtime pay of 5 employees. Overtime is paid at the rate of Rs. 12 per hour for every hour worked above 40 hours.(Use array of object)
- 5) Write a program to implement cricket class to accept and display 5 records of cricketers with appropriate data members.
- 6) Write any program which shows use of inline function.
- 7) Write a program which shows passing object as parameter.
- 8) Write a program which shows the function overloading concept.
- 9) Write a program which shows the nested class concept.
- 10) Write a program which accept 10 numbers dynamically and find minimum number from them. (Use new operator)
- 11) Write a program which shows the concept of creating dynamic object.(Use new operator)
- 12) Write a program to implement 'book' class with appropriate data members and get () as member function. Accept all values of data members in get (). And display all accepted values in friend function show().
- 13) Write a program which shows friend function for two classes.
- 14) Write a program which shows the use of friend class.

ASSIGNMENT-4

- 1) Write a program to implement default constructor.
- 2) Write a program to implement parameterized constructor.
- 3) Write a program to implement copy constructor.
- 4) Write a program for constructor overloading. (Multiple constructor in a class)

Que: Write following programs by using Constructor and Destructor.

- 1) Find the largest and smallest number from the array.
- 2) Swap two numbers by using parameterized constructor.
- 3) Check entered string is Palindrome or not.
- 4) Find the sum of diagonal elements of the matrix.
- 5) Print the Fibonacci series up to 'n' numbers.

ASSIGNMENT-5

- 1) Write a program to overload +, -, *, / operators by using member function of the class.
- 2) Write a program to overload +, -, *, / operators by using friend function.
- 3) Write a program to overload increment (+ +) and decrement (--)
operator.
- 4) Write a program to overload the relational operators < and > .
- 5) Write a program to overload the assignment (=) operator.
- 6) Write a program to overload the logical operator (&&) to check
maximum number between two numbers.
- 7) Write a program to overload + operator for concatenation of two string.
- 8) Write a program to find simple interest of entered principal (P), rate of
interest (R) and number of year (N) using operator overloading.
- 9) Write a program using operator overloading to find out the factorial of
the number.